# BALOCHISTAN BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, QUETTA

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2024

# MATHEMATICS

Past Paper 2024

Retyped faithfully from the original board paper — same questions, no changes.

## PART - I

Note: Q.1 Choose the correct answer. (12x1=12)

Note: NOTE: Part-I is compulsory. Attempt eight questions from Part-II, five questions from Part-III and two questions from Part-IV. Only simple calculator is allowed.

Q1. Sum of the cube roots of unity is -----.

(A) 0 (B) 1

(C) 2 (D) 3

Q2. If y varies directly as x then it can be written as -----.

(A) $y \propto 1/x$ (B) $y \propto x$

(C) $K \propto x$ (D) $y \propto 1/K$

Q3. $(x-1)/((x+7)(x^2+9))$ can be decomposed in partial fraction -----.

(A) $A/(x+7) + B/(x^2+9)$ (B) $(Ax+B)/(x^2+9) + C/(x+7)$

(C) $A/(x+7) + (Bx+C)/(x^2+9)$ (D) $A/(x+7) - (Bx+C)/(x^2+9)$

Q4. $(A \cup B)^c$ = -----.

(A) $A \cup B$ (B) $A \cap B^c$

(C) $A^c \cup B^c$ (D) $A^c \cap B^c$

Q5. Histogram is a graph of -----.

(A) Frequency (B) Class limit

(C) Ungrouped data (D) Frequency distribution

Q6. An angle is a union of two -----.

(A) Non collinear rays (B) Collinear rays

(C) Initial rays (D) Terminal rays

Q7. An angle inscribed in a semi circle is a ----- angle

(A) Area (B) Sector

(C) Right (D) Chord

Q8. If $B \subseteq A$ then $A \cup B$ = -----.

(A) B (B) $B^c$

(C) $A^c$ (D) A

Q9. Third proportional of 9 and 6 is -----.

(A) 4 (B) 6

(C) 12 (D) 15

Q10. Sum of roots of quadratic equation $2x^2 + 3x + 7 = 0$ is -----.

(A) -3 (B) -3/2

(C) 1/2 (D) 3/2

Q11. The tangent and radius of a circle at the point of contact are -----.

(A) Parallel (B) non parallel

(C) perpendicular (D) Double

Q12. A chord passing through the center of circle is called -----.

(A) Radius (B) Collinear

(C) Diameter (D) Medium

## PART - II

> Note: Note:- Attempt any eight (8) questions. (8x3=24)

(i) Solve by factorization $3x^2 - 12x(x - 1) = 6$

(ii) Discuss the nature of roots $3x^2 + 2x - 4 = 0$

(iii) Prove that $1 + W^{34} + W^{47} = 0$

(iv) Find a if 40Kg : 90Kg :: 360 m : a m.

(v) If U = {1,2,3,4,.....40}, X = {21,23,25 ..... 39} and Y = {22,24,26 ..... 40} then find $(X - Y)^c$

(vi) Use synthetic division find value of P if the zero of polynomial $5x^3 + 8x^2 - 7x + 8P$ is +2.

(vii) Find the geometric mean of the observation 4,16 and 64 by using basic formula.

(viii) A sector of a circle of radius 12cm has an area of $128cm^2$. Find angle subtended at center of the circle.

(ix) Find K, the roots of $Kx(x - 2) + 6 = 0$ are equal.

(x) Define Radical equation and Histogram.

## PART - III

> Note: Note:- Attempt any five (5) questions. (5x5=25)

Q3. (Marks 5) Solve by completing square $2x^2 + 8x + 4 = 0$

Q4. (Marks 5) Solve $\sqrt{(x+3)} + \sqrt{(x+8)} = \sqrt{(x+24)}$

Q5. (Marks 5) The sum of square of two consecutive odd integers is 34. Find the integers.

Q6. (Marks 5) Resolve into partial fractions $(2x+7)/(x^2-4)$

Q7. (Marks 5) Find arithmetic mean by direct method and indirect method for the following data 15, 29, 25, 18, 45, 34, 39, 32, 41, 26

Q8. (Marks 5) Show that $1/(\sec\theta - \tan\theta) = \sec\theta + \tan\theta$

Q9. (Marks 5) Solve the following simultaneous equations. $2x^2 = 6 + 3y^2$, $3x^2 - 5y^2 = 7$

Q10. (Marks 5) If U:V = 4:7, find the value of 3U + 5V : 6U - 5V

## PART - IV

> Note: Note:- Attempt any two (2) questions. (2x7=14)

Q11. (Marks 7) One and only one circle can pass through three non collinear points (Prove).

Q12. (Marks 7) The opposite angles of any quadrilateral inscribed in a circle are supplementary (Prove).

Q13. (Marks 7) The area enclosed between two concentric circle in $770cm^2$, radius of the outer circle is 21cm. Find the radius of the inner circle.

Q14. (Marks 7) Draw a tangent to a circle with radius 3.5cm at point P on the circumference.

> Note: THE END